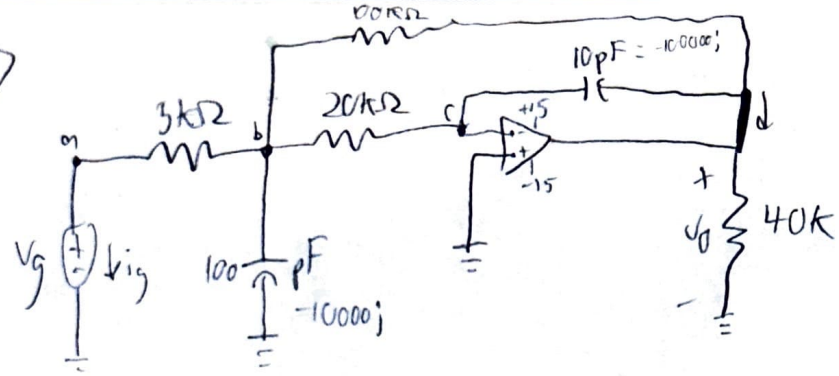
9.67



$$\frac{V_a - V_b}{5000} + i_g = 0$$

$$\frac{V_b - V_a}{5000} + \frac{V_b}{-10000j} + \frac{V_b - V_d}{100000} + \frac{V_b - V_c}{20000} = 0$$

$$\frac{V_c - V_b}{20000} + \frac{V_c - V_d}{-100000j} = 0$$

$$\frac{V_d - V_c}{-100000j} + \frac{V_d - V_b}{100000} + \frac{V_d}{40000} = 0$$

$V_d = V_o$ $\quad V_a = V_g = 2\angle 0$

$V_c = 0$

mattlab: $V_a = 2$

$V_b = \frac{1040}{201} - \frac{200j}{201}$

$V_d = \frac{1000 + 5200j}{201}$

$V_d = V_o$

$1.43 + 7.42j$

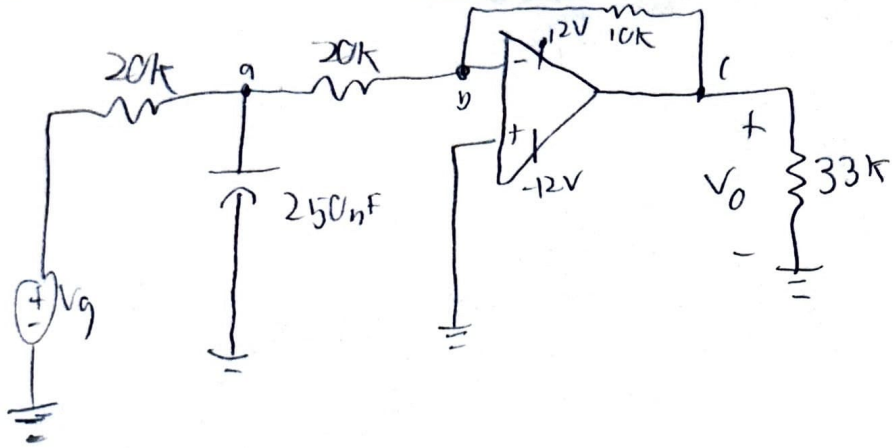
$\sqrt{1.43^2 + 7.42^2} = 7.55$

$\tan^{-1}\left(\frac{7.42}{1.43}\right) = 79.11^0$

$$\boxed{V_o(t) = 7.55\cos(10^6 t + 79.11)\,V}$$

✓ less than 15

9.69



$$\frac{V_a - V_g}{20k} + \frac{V_a}{-20000j} + \frac{V_a}{20k} = 0$$

$$\frac{-V_a}{20000} + \frac{-V_c}{10000} = 0$$

$$V_a = \frac{8}{5} - \frac{4j}{5}$$

$$V_c = -\frac{4}{5} + \frac{2j}{5}$$

$\tan^{-1}\left(\frac{\frac{2}{5}}{-\frac{4}{5}}\right) = -26.565$ but are in the 2nd quad so 153.43^0

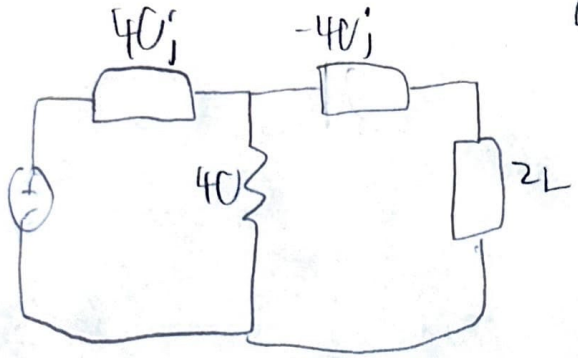
$$V_c = .89 \angle 153.43^0$$

$$V_c = V_o$$

$$\boxed{V_o(t) = .89\cos(200t + 153.43)\ V}$$

10.44

$80\cos 5000t$

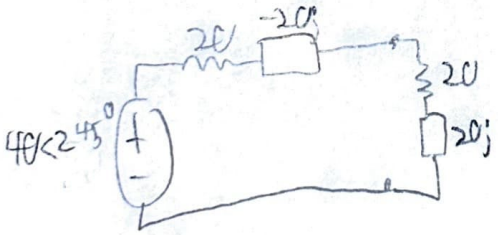


$$\frac{1600j}{40+40j} = 20+20j$$

$$20+20j - 40j$$

$$20-20j = Z_{th}$$

$$Z_L = Z_{th}^* = \boxed{20+20j}$$ 9.



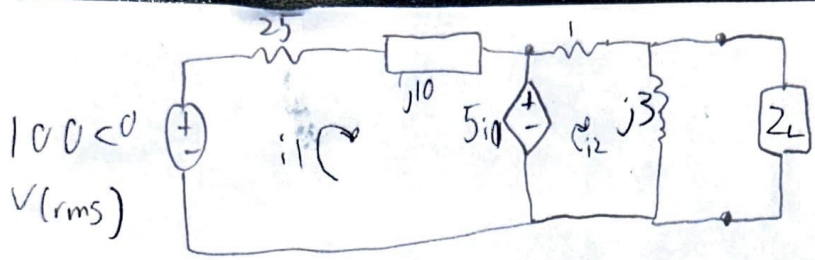
$$80\left(\frac{20+20j}{40}\right) = 40+40j \checkmark$$

$$I_{rms} = \frac{40\sqrt{2}\angle 45}{\sqrt{2}\,(20+j20-20j+20)} = 1\angle 45$$

$$\frac{1}{2}(1^2+1^2)20) = 20$$

20 watts

10.46



$-100 + i1(25+j10) + 5i\phi = 0$

$-5i\phi + i2(1+j3) = 0$

$-15+5j + i2(1+j3) = 0$

$i2(1+j3) = 15-5j$

$i2 = -5j$

$-5j \cdot j3 = 15$

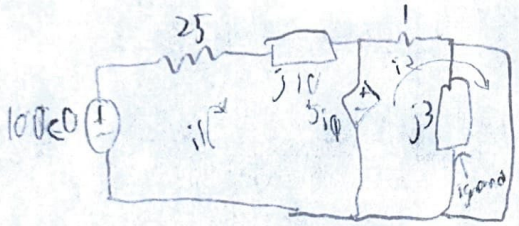
$V_{th} = 15$

$5i\phi = \frac{100-5i\phi}{25+j10}(5)$

$(25+j10)i\phi = 100 - 5i\phi$

$i\phi((25+j10)+5) = 100$

$i\phi = \frac{100}{30+j10} = 3-j$



$-100 + i1(25+j10) + 5\phi = 0$

$-100 + i1(25+j10) + 5(3-j) = 0$

$-5i\phi + i2 = 0$

$i2 = 5i\phi$

$i2 = 15-5j$

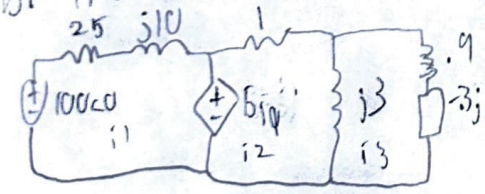
$isc = 15-5j$

$\frac{15}{15-5j} = .9+.3j$

$Z_{th} =$

$Z_{th}^* = Z_L$

b. circuit



$-100 + i1(25+j10) + 5i\phi = 0$

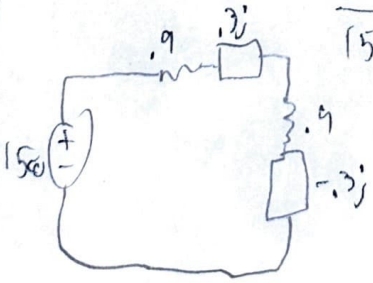
$i1 = i\phi$

$-100 + i1(25+j10) + 5i1 = 0$

$30i1 + i1j10 = 100$

$|(3-j)^2| = 10$ $i1 = 3-j$

b. 16.67Ω



$\frac{15}{1.8} = 8.33A$

a. $(8.33^2)(.9) = 62.5W$

$-5i\phi + i2 + (i2-i3)j3 = 0$

$-15+5j + i2 + i2j3 - 8.33(j3) = 0$

$i2 = (-4.5-1.5j)$

$(-4.5-1.5j)(15-5j) = -75$

$-100(3+j) = -300-100j$

$-300-75 = -375$

$i3 = 8.33$

$\frac{62.5}{375} = 16.6\overline{6}$